

2 Three new directions share one nested WPBE design

Takeaway. Starting from the two-page benchmark, we change only second-window supply: retained incumbents, time-homogeneous fresh arrivals, and finally an expanded rescue footprint. Prices and search are always optimized outside the induced cutoff-WPBE.

Architecture	Policy	Who can volunteer in window 2	What it identifies
I: retained incumbents	(p_1, p_2)	First-window rejectors who remain available	Same-order recall value
A: fixed footprint + arrivals	(p_1, p_2)	Retained rejectors plus an independent fresh core cohort	Time-homogeneous replenishment
E: expanded rescue search	(p_1, p_2, s)	The same core supply plus an outer annulus reached only in rescue	Incremental geographic reach

Here p_j is one anonymous transaction price: the rider pays p_j and the assigned driver receives p_j . The local per-window flows are

$$N_1 \sim \text{Pois}(m), \quad N_2^C \sim \text{Pois}(m), \quad N_2^O(s) \sim \text{Pois}(m(s-1)). \quad (5)$$

Conditional on first-window cutoff a , define retained-incumbent, fresh-core, and outer-annulus volunteer intensities by

$$I_j(a) = \omega m[F(p_j) - F(a)]_+, \quad E_j^C = mF(p_j), \quad E_2^O(s) = m \int_1^s F(p_2 - \tau(\sqrt{u} - 1)) du. \quad (6)$$

The total terminal intensity is

$$\lambda_j^r(a) = I_j(a) + \mathbf{1}\{r \neq \text{I}\} E_j^C + \mathbf{1}\{r = \text{E}, j = 2\} E_2^O(s), \quad C_j^r = 1 - e^{-\lambda_j^r}, \quad h(\lambda) = \frac{1 - e^{-\lambda}}{\lambda}. \quad (7)$$

Thus every λ is an expected Poisson volunteer count, C is coverage, and h is one volunteer's lottery win probability. Only $\text{A} \subset \text{E}$: $s = 1$ reproduces the fixed footprint, while the rescue radius is $R_0\sqrt{s}$.

Information and incentives. Incumbents have seen the order and commitment, so they may reject and wait. Fresh drivers first see it in window 2. Universal rejection is public; the rider observes failure but not realized counts, and uses equilibrium coverage. A rejector remains eligible with probability ω .

Two different costs. An assigned outer driver pays the post-assignment pickup cost embedded in (6). Separately, after failure the platform reserves capacity for its promised outer footprint:

$$Q^{\text{com}} = (1 - p_1)e^{-ma}m(s-1). \quad (8)$$

Thus free search is not built into the objective.

Every plotted policy is evaluated at its own cutoff-WPBE:

$$V_r(\theta) = \max_{\mu_r} J_r(\mu_r, a^*(\mu_r; \theta)), \quad J_r = M_r - \kappa Q_r^{\text{com}}, \quad \boxed{\text{fix } p_1 \rightarrow \max_{p_2, s} \rightarrow \max_{p_1}}. \quad (9)$$

The inner cutoff a^* , rider continuation, and supply composition are all re-solved at every outer-policy evaluation; s is present only for E.

3 Directions 1–2: isolate the value of fresh arrivals

Takeaway. Adding a fresh core cohort has its largest displayed value in a moderately thick market; the option to search outside the core is most valuable earlier, when the local market is still thin.

Fresh replenishment matters later; geographic reach matters first

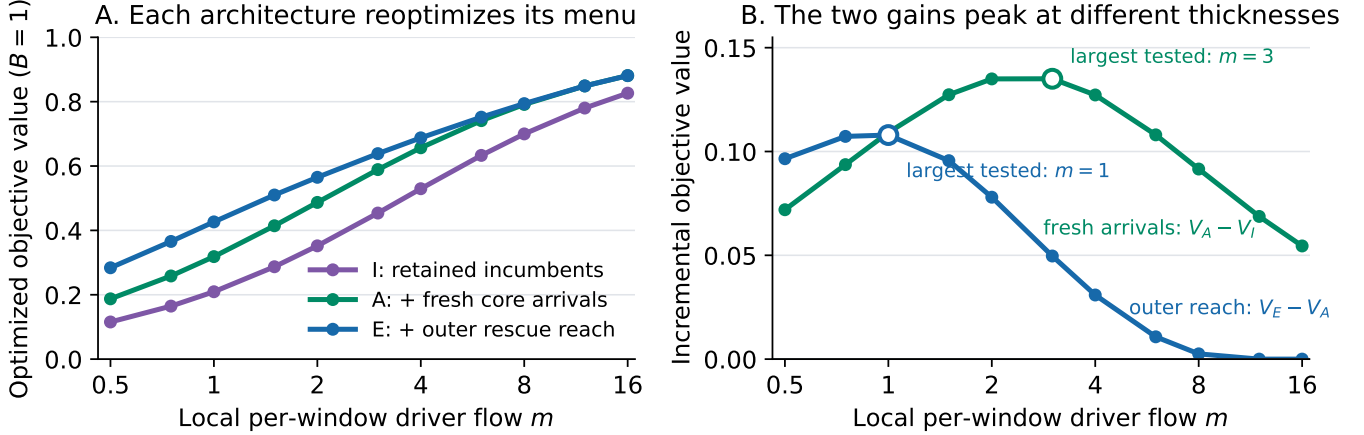


Figure 1: Illustrative slice $(\beta, \delta, \omega, \tau, \kappa, \bar{s}) = (.8, .8, .8, .25, .0125, 4)$ with $v, c \sim U[0, 1]$ and $B = 1$. Every point separately reoptimizes all controls available to that architecture and re-solves the cutoff-WPBE. I and A report completion; E reports completion net of committed-reach cost.

- **I $\rightarrow$ A: replenishment.** Fresh arrivals directly raise second-window coverage and also compete with rejectors who wait. The gain is therefore an equilibrium response, not a fixed-policy supply addition.
- **A $\rightarrow$ E: reach option.** Because $s = 1$ is feasible, this net option value is nonnegative. Its largest tested value occurs around $m = 1$, before the fresh-core value reaches its largest tested value around $m = 3$.
- **Interpret the peaks cautiously.** The plot supports two distinct economic regions and an ordered-peak conjecture; it does not claim unique global peaks in the fully endogenous WPBE model.

4 Direction 3: expanded search is an early-thickness tool

Takeaway. Market thickness changes the mix of recovery tools. Search scope contracts sharply with m , but a positive rescue price increment remains even after the optimized footprint returns to the core.

Search scope contracts before the rescue price increment disappears

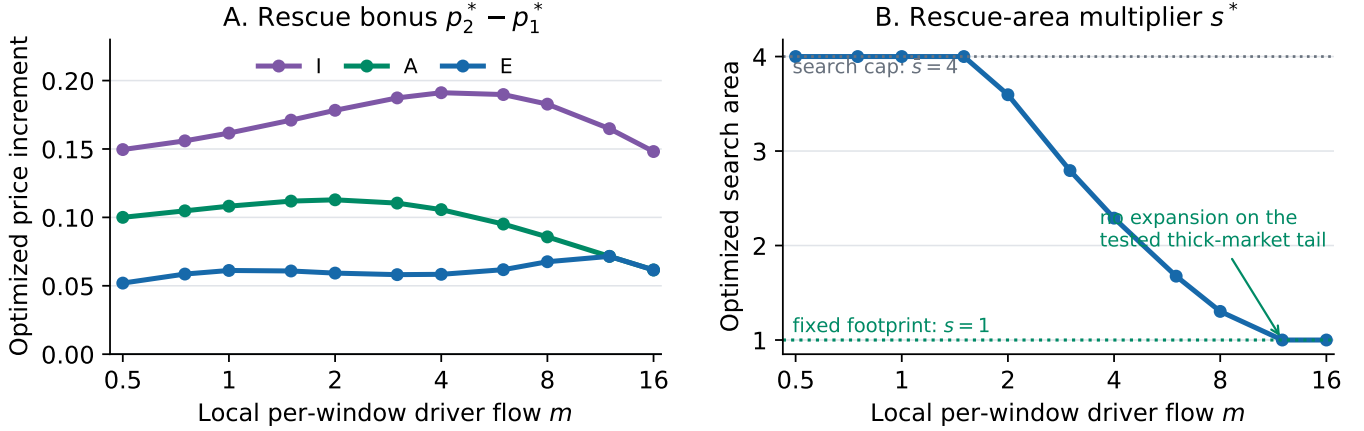


Figure 2: Same calibration and equilibrium-constrained optimization as page 4. The right panel uses committed-reach cost; $s = 1$ means no expansion beyond the fresh core cohort.

Market region	Main shortage	Numerical mechanism intuition
Thin	Too few reachable drivers	Expand aggressively; the tested optimum reaches the search cap.
Intermediate	Coverage and strategic waiting both matter	Combine a rescue increment with a gradually shrinking outer footprint.
Thick	The core is already reliable	Set $s = 1$ on the tested tail, while retaining a positive $p_2 - p_1$.

Current story. Fresh arrivals and expanded search are different sources of recovery value. Search is an early-thickness tool; rescue pricing continues to manage screening and delayed acceptance after geographic expansion is no longer worth its capacity cost.

The next theory step is deliberately deferred: primitive conditions for peak ordering and the exact no-search threshold in κ . The complete theory note keeps those claims, proofs, counterexamples, and literature boundaries separate from this five-page intuition-and-simulation presentation.